

Matrices

Multiplication of Matrices

- Recall scalar multiplication
- Given a matrix C, if the matrix is multiply by a number, them all the elements in the matrix is multiply by the number.

$$C = \begin{bmatrix} 4 & 7 & 12 \\ 6 & 9 & 16 \end{bmatrix}$$

$$2C = \begin{bmatrix} 2 \times 4 & 2 \times 7 & 2 \times 12 \\ 2 \times 6 & 2 \times 9 & 2 \times 16 \end{bmatrix} = \begin{bmatrix} 8 & 14 & 24 \\ 12 & 18 & 32 \end{bmatrix}$$

Matrix multiplication

- Consider the following matrices:

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 4 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & -4 \\ 3 & 8 \\ -10 & 9 \end{bmatrix}$$

$$A \times B = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 4 & 5 \end{bmatrix} \times \begin{bmatrix} 1 & -4 \\ 3 & 8 \\ -10 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \times 1 + 1 \times 3 + 3 \times -10 & \end{bmatrix}$$

$$= \begin{bmatrix} -25 & \end{bmatrix}$$

$$A \times B = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 4 & 5 \end{bmatrix} \times \begin{bmatrix} 1 & -4 \\ 3 & 8 \\ -10 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} -25 & 2 \times -4 + 1 \times 8 + 3 \times 9 \end{bmatrix}$$

$$= \begin{bmatrix} -25 & 27 \end{bmatrix}$$

$$A \times B = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 4 & 5 \end{bmatrix} \times \begin{bmatrix} 1 & -4 \\ 3 & 8 \\ -10 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} -25 & 27 \\ 1 \times 1 + 4 \times 3 + 5 \times -10 & \end{bmatrix}$$

$$= \begin{bmatrix} -25 & 27 \\ 37 & \end{bmatrix}$$

$$A \times B = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 4 & 5 \end{bmatrix} \times \begin{bmatrix} 1 & -4 \\ 3 & 8 \\ -10 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} -25 & 27 \\ -37 & 1 \times -4 + 4 \times 8 + 5 \times 9 \end{bmatrix}$$

$$= \begin{bmatrix} -25 & 27 \\ -37 & 73 \end{bmatrix}$$

$$A \times B = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 4 & 5 \end{bmatrix} \times \begin{bmatrix} 1 & -4 \\ 3 & 8 \\ -10 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \times 1 + 1 \times 3 + 3 \times -10 & 2 \times -4 + 1 \times 8 + 3 \times 9 \\ 1 \times 1 + 4 \times 3 + 5 \times -10 & 1 \times -4 + 4 \times 8 + 5 \times 9 \end{bmatrix}$$

$$= \begin{bmatrix} -25 & 27 \\ -37 & 73 \end{bmatrix}$$

Other examples

$$\begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix},$$

$$\begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} -1 & 13 \end{bmatrix}$$

1×2
 2×2
 1×2

$$\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 17 \\ 1 & 2 \end{bmatrix}$$

2×2
 2×2
 2×2

$$\begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 6 & 4 \end{bmatrix}$$

1×2
 2×3
 1×3

$$\begin{bmatrix} 3 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 9 \\ 16 \end{bmatrix}$$

2×2
 2×1
 2×1

$$\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 17 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 11 \\ -1 & 9 \end{bmatrix}$$

In matrix multiplication, $AB \neq BA$, therefore matrix multiplication is not commutative.

Two matrices can be multiplied together if the number of columns in the first matrix equals the number of the number of rows in the second matrix.

If matrix A has dimensions of $m \times n$ and matrix B has dimensions of $p \times q$

- The matrix product $A_{m \times n} B_{p \times q}$ can only be formed if $n = p$ and AB will have dimensions $m \times q$
- The matrix product $B_{p \times q} A_{m \times n}$ can only be formed if $q = m$ and AB will have dimensions $p \times n$

In the product AB,

- B is *premultiplied* by A
- A is *postmultiplied* by B

Practical examples

	1 st Place	2 nd Place	3 rd Place
School A	1	1	1
School B	3	1	0
School C	0	3	3
School D	1	2	0
School E	2	0	3

1st	3 points
2nd	2 points
3rd	1 point

$$\begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & 0 \\ 0 & 3 & 3 \\ 1 & 2 & 0 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \\ 9 \\ 7 \\ 9 \end{bmatrix}$$

A manufacturer makes three products A, B and C, each requiring a certain number of units of commodities P, Q, R, S and T. Matrix X below shows the number of units of each commodity required to make one of each product.

$$\begin{array}{l} \text{Product A} \\ \text{Product B} \\ \text{Product C} \end{array} \begin{bmatrix} \text{P} & \text{Q} & \text{R} & \text{S} & \text{T} \\ 1 & 1 & 0 & 2 & 3 \\ 1 & 1 & 2 & 1 & 2 \\ 0 & 1 & 3 & 0 & 3 \end{bmatrix} = X$$

- (a) Each unit of P, Q, R, S and T costs the manufacturer \$200, \$100, \$50, \$400 and \$300 respectively. Write this information as matrix Y which should be either a column matrix or a row matrix, whichever can form a product with X.
- (b) Form the product referred to in (a) and explain what information it displays.

Zero matrices

- All elements in a zero matrix is 0.
- The letter O is used to represent a zero matrix

- For any matrix A,

- $A + O = A$
- $O + A = A$
- $A * O = O$
- $O * A = O$

O is referred to
as *additive
identity matrix*

Note

- If x and y represents number, if $xy = 0$, then either $x = 0$ and/or $y = 0$.
- However for matrices, if $AB = O$, it is not necessarily that A and/or $B = O$

$$\begin{aligned} AB &= \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

Note

- When x , y and z represent numbers, if $xz = yz$ and $z \neq 0$, then $x = y$.
- But for matrices, if $AB = CB$ and $B \neq O$, matrix A is not necessarily the same as matrix C .

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} -3 & 2 \\ 5 & 2 \end{bmatrix}.$$

$$AB = \begin{bmatrix} 3 & -1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \end{bmatrix}$$

$$CB = \begin{bmatrix} -3 & 2 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \end{bmatrix}.$$

Multiplicative Identity Matrices

- A multiplicative identity matrix leaves all other matrices unchanged under multiplication
- If $I_{m \times n}$ is a multiplicative matrix
 - $I_{m \times n} A_{n \times p} = A_{n \times p}$ thus $m = n$
 - $B_{q \times m} I_{m \times n} = B_{q \times n}$ thus $m = n$
- Therefore I is a square matrix.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Note

- In numbers, if $xy = x$, for $x \neq 0$, then $y = 1$
- But for matrices $AB = B$, $B \neq 0$, A is not necessarily $= 1$.

$$\begin{bmatrix} 4 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix}$$